

Statement of Objectives
SATCOM Modernization
(Deployable Dual-Hub SATCOM Terminal)

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Statement of Objectives (SOO) - Replacement of Dual-Hub SATCOM Apprentice Training Terminals

1.0 Purpose

This Statement of Objectives (SOO) establishes the performance objectives for the replacement of two (2) end-of-life Ground Multi-Band Terminal (GMT) and Quad-band Large Aperture Antenna (QLAA) training assets. The primary purpose of this acquisition is to sustain the apprentice-level training mission by replacing legacy equipment with modern, efficient, and sustainable systems. This is a like-for-like replacement intended to maintain existing training capabilities and is not the acquisition of a new capability.

2.0 Scope

The scope of this SOO includes all necessary hardware, software, integration, documentation, and training required to deliver two complete, turn-key, and fully operational deployable dual-hub SATCOM terminal systems to replace the Government's existing assets.

3.0 Background

The existing GMT and QLAA systems are beyond their planned service life. They suffer from frequent downtime, require excessive maintenance hours, and lack the operational efficiency of modern designs. This situation creates risks to the training pipeline and incurs unnecessary costs in maintenance and manpower. A modern replacement is required to ensure the continued, uninterrupted training of apprentice-level personnel in a more cost-effective manner.

4.0 Performance Objectives

The contractor shall provide a solution that meets the following high-level performance objectives:

4.1 Objective 1: Sustain Multi-Band, Dual-Hub Training Operations

The contractor shall deliver two SATCOM terminals that sustain existing training operations utilizing two small aperture (1.2-2.6m) antennas per terminal, with each terminal supporting two independent communication links (dual-hub) simultaneously. The systems must operate across C, X, Ku, and Ka-band frequencies. To reduce maintenance complexity, this shall be accomplished via a modular feed boom design that allows for simple, tool-less swapping between frequency bands.

4.2 Objective 2: Reduce Manpower and Increase Operational Efficiency

To reduce instructor and student manpower overhead and increase training efficiency, the system must achieve the following outcomes:

- **Be deployable and fully operational by a two-person crew.** This includes all setup, satellite acquisition, and teardown procedures.
- **Support full remote command and control** from a classroom environment via two contractor-provided ruggedized laptops with intuitive GUIs per terminal. This capability is critical to minimizing manual intervention and reducing equipment wear.
- **Automate satellite acquisition** with motorized drives to reduce the time and skill level required to establish a link during training exercises.
- **Operate on a low-power amplifier configuration** sufficient for SATSIM operations, reducing lifecycle energy costs and thermal load.

4.3 Objective 3: Ensure Long-Term Durability and Organic Sustainment

To ensure the long-term service life of the equipment and support an organic maintenance posture, the system design must:

- **Be fully containerized** in ruggedized, transportable transit cases for all components to protect them during storage and movement.
- **Integrate seamlessly with existing infrastructure**, including mounting on standard concrete pads (rated for 65 MPH wind gusts) and connecting to 120/240 VAC power sources.
- **Facilitate organic maintenance** through a modular design that allows for simple component replacement (minimum: feed horns and modems).

4.4 Objective 4: Integrate with Existing Training Curriculum

The system shall seamlessly integrate with the Government's existing curriculum. This includes:

- Providing dual-chain modem assemblies that support current and legacy configurations (minimum: BPSK and QPSK) and interface standards (minimum: MIL-STD-188-165A and EIA-530) to ensure compatibility with established training scenarios (i.e., iDirect).
- Providing all necessary indoor equipment (laptops, UPS) and interfacing links required for operation from the classroom.

5.0 Key Deliverables

To meet the objectives of this SOO, the contractor shall provide the following key deliverables:

- **Two (2) complete Dual-Hub SATCOM Terminal Systems:** Fully functional and meeting all objectives defined in Section 4.0.
- **On-Site Integration:** A turn-key set-up and integration service to ensure the systems are fully operational within the existing classroom environments.
- **Comprehensive Technical Documentation:** All user/technical manuals, schematics, and maintenance guides sufficient for organic, operator-level and intermediate-level maintenance, eliminating the need for future vendor support contracts.

- **One-Time "Train-the-Trainer" Program:** A single, comprehensive training event for up to 12 government personnel to establish a self-sufficient, in-house training capability.

6.0 Place of Performance

- All equipment components shall be delivered to 809 Hercules St (Bldg 6902, Jones Hall), KAFB, MS 39534.

7.0 Period of Performance

- The contractor shall deliver, set-up, and complete all training and integration activities between 180 and 225 days after contract award.